FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Main)-2015

ANSWERS, HINTS & SOLUTIONS
FULL TEST – I
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	A	B
2.	B	B	A
3.	A	C	C
4.	A	A	D
5.	A	D	C
6.	A	B	A
7.	A	D	B
8.	A	C	B
9.	B	B	C
10.	D	A	A
11.	D	D	C
12.	D	B	B
13.	A	D	C
14.	C	A	C
15.	D	C	A
16.	C	B	A
17.	A	A	C
18.	C	B	B
19.	A	B	A
20.	A	B	D
21.	B	D	C
22.	B	D	C
23.	B	C	B
24.	D	D	C
25.	A	D	C
26.	D	D	A
27.	D	B	D
28.	C	A	A
29.	A	D	B
30.	A	A	B

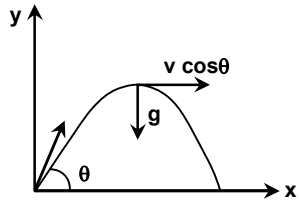
Physics

PART – I

SECTION – A

1. Rate of change of speed is minimum at highest point. Since at highest position.

$$R = \frac{(v\cos\theta)^2}{g} = \frac{v^2\cos^2\theta}{g}$$



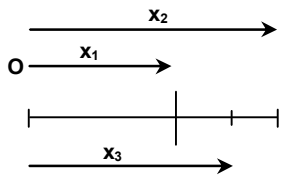
2. $x = t^3 - 4t^2 + 5t + 9$

$$V = \frac{dx}{dt} = (3t^2 - 8t + 5)$$

V = 0 at t = 1, $\frac{5}{3}$ sec.

x_1 (t = 0) = 9m; x_2 (t = 1) = 11 m, x_3 (t = 5/3) ≈ 10.85

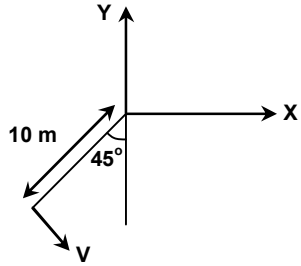
d = $(x_2 - x_2) + (x_2 - x_3)$ = 2.15 m



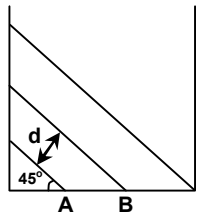
3. $$\frac{d|\vec{v}|}{dt} = a_t = \frac{\vec{a}.\vec{v}}{|\vec{v}|} = \frac{24}{5} = 4.8\ ms^{-2}$$

4. Situation is same as the motion in vertical circle with initial situation |a| = 2 equivalent to lowest point.

$$V = \sqrt{5a\ell} = \sqrt{5\times 2\times 10} = 10\,ms^{-1}$$



5. $$|P_A - P_B| = d \times g_{effective} \times \rho = 10^4\ N/m^2$$



6. Friction would change the diagram with time as tension in string would increase. So, (a) is correct option.

7. $V_B \cos\theta_1 = 10$; $v_B \cos\theta_2 = v_c$; $v_c = \frac{10}{\cos\theta_1}\cos\theta_2 = \frac{10}{\sqrt{3}}$ m/sec.

8. Net force would be equal to figure mg when contact force with hemispherical ball is zero. $\theta = \cos^{-1}(2/3)$

9. Angle between $\vec{V}$ & $\vec{a}$ is obtuse and centripetal acceleration is directed towards left.

10. I = 10 tan (90 – θ)

$$\frac{dI}{dt} = 10\sec^2\theta \times \omega$$

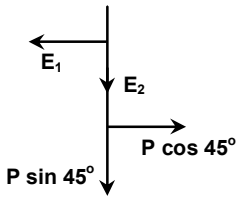
= 500 m/sec

11. $$\frac{\sin i}{\sin r} = \frac{\sqrt{5}}{2\sqrt{2}}$$

$$\left[\sin r = \frac{4}{5}\right]$$

12. $$E_1 = -\frac{KP\cos 45^\circ}{Y^3}$$

$$E_2 = \frac{2KP\sin 45^\circ}{Y^3}$$



13. $$V_0 - \frac{\sigma\ell}{\varepsilon_0} + \frac{q}{4\pi\varepsilon_0 R} = 0$$

14. Initially capacitor will offer zero resistance at steady state it will behave like open circuit.

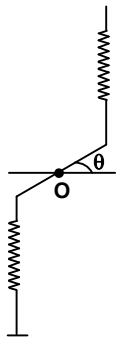
15. By KVL & KCL

16. $$\frac{dN_y}{dt} = \lambda_x N_x - \lambda_y N_y$$

17. $$X = 2T\cos\theta$$

$$Y = 2\tan\theta$$

$$\frac{X}{Y} = \frac{(R-d)}{R}$$



18. Torque about O

$$-(\lambda/2)\,(\lambda/2)\,\theta + k\,(\lambda/2)\,(\lambda/2)\,\theta = \frac{M\ell^2}{12}\alpha$$

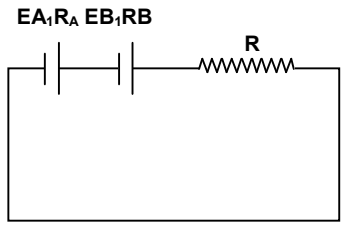
$$\frac{K}{2}\theta = \frac{M\ell^2}{612}\alpha$$

$$\omega^2 = \frac{6K}{M}$$

$$T = 2\pi\sqrt{\frac{M}{6K}}$$

19. $T = 2\pi\sqrt{\frac{\ell}{g\cos\alpha}}$

20. At steady state current through capacitor is zero and potential difference across inductor is zero. Circuit will be like $i = \frac{E}{r_A}$.



$2r_A = r_A + r_B + R$

$R = r_A - r_B$ since R > 0.

21. $B = \frac{\mu_0 i}{2\pi x}$

$d\varepsilon = Bd\ell v$

$\varepsilon = \frac{\mu_0 iv}{2\pi}\int_{r_2}^{r_1}\frac{dx}{x} = \left(\frac{\mu_0 iv}{2\pi}\ell n\frac{r_2}{r_1}\right)$

22. $\left[\begin{array}{ll}\frac{hc}{\lambda} & = \omega_f + ev \\ \frac{hc}{2\lambda} & = \omega_f + \frac{eV}{3}\end{array}\right]$

$\omega_f = \frac{hc}{4\lambda}, \lambda_0 = 4\lambda$

23. r particle will go undefected since it does not have α, β particle will deflect in opposite directions

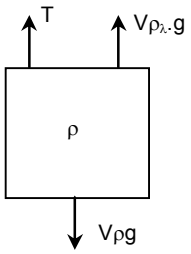
24. $X = \lambda N_1$

$Y = \lambda N_2$

$(N_1 - N_2) = \frac{(x-y)}{\lambda} = (x-y)\tau$

25. $T + V\rho_\lambda.g = 40$

So T = 37.12 N



26. $V_2 = 2V_1; K_2 = 4K_1$

27. $|E| = (aT - 2at)$

$H = \int E^2/R\,dt$

28. $-\frac{dh}{dt} \propto \sqrt{h}$

29. $E = BV\ell = 80\,V$

30. Condition of balanced bridge.

Chemistry

PART – II

SECTION – A

1. $\pi P_c = P \qquad \phi V_c = V \qquad \theta T_c = T$

$\pi.\frac{a}{27b^2} = P \qquad \phi.3b = V \qquad \theta.\frac{8a}{27Rb} = T$

$\left[\pi - \frac{3}{\phi}\right](3\phi - 1) = 8\theta R$

4. $[H_2CO_3] = 2M$
$V_{blood} = 10$ ml
$[NaHCO_3] = 5$ M
Let V ml 8 $NaHCO_3$ is mixed
$\therefore$ total V = (10 + V) ml

$\therefore [H_2CO_3]_{mix} = \frac{2\times 10}{(v+10)}$

$[NaHCO_3] = \frac{5\times V}{(v+10)}$

$\therefore$ pH = pKa + $\log\frac{[salt]}{[acid]}$

$\Rightarrow 7.4 = -\log(7.8\times 10^{-7}) + \log\frac{5V}{20}$

$\Rightarrow$ V = 78.36 ml

8. For a chemical reaction with rise in temperature by 10°, the rate of reaction is nearly doubled. The temperature dependence of the rate of a chemical equation can be explained by Arrhenius equation:
$K = Ae^{-E/RT}$

18. Number of G.I. = $2^4 = 2^1 = 2$
Number of OI = $2^4 = 2^1 = 2$
n = number of stereocentres

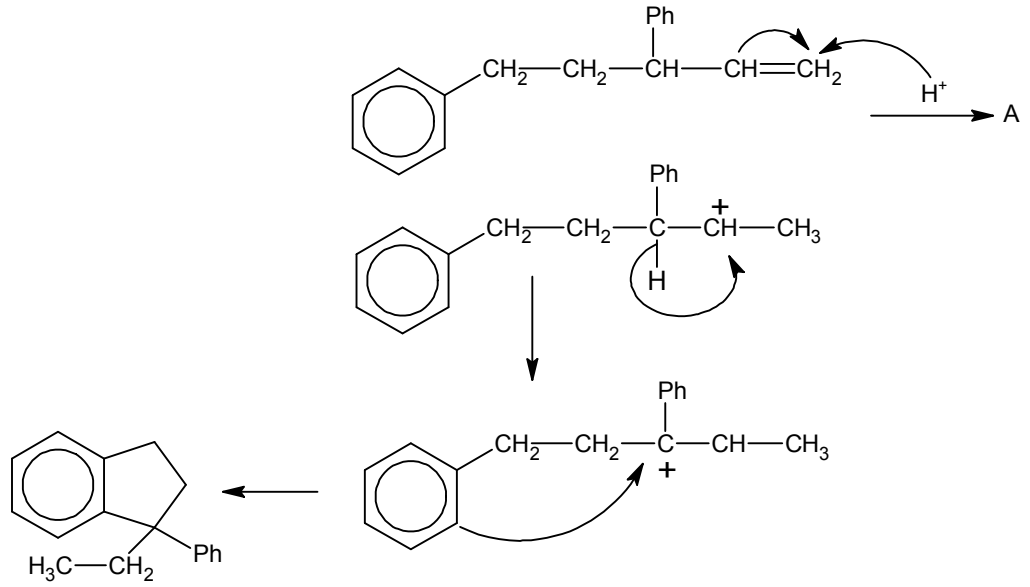
28. $P = \sqrt{2\times 9.11\times 10^{-31}(KJ)\times 1.602\times 0^{-17}}$

$= 5.4\times 10^{-24}$ kgms^{-1}

$\lambda = \frac{h}{p} = \frac{6.6\times 10^{-31}JS}{5.4\times 10^{-24}\,kgms^{-1}} = 0.1226$

29. Bond length $\alpha \frac{1}{\%\pi \text{ characater}}$

30.



Mathematics PART – III

SECTION – A

1. For x > 1, let $f(x) = 5^{\log_{10}(x^3 - x^2)}$

The there equation are $f(x) = y^2$, $f(y) = z^2$ and $f(z) = x^2$

$\because$ $x^3 - x^2 = x^2(x - 1)$ is increasing, f is an increasing function

If, say x > y then y < z and z < x, yielding a contradiction

Thus we can only have that x = y = z and so $\log_{10}(x^3 - x^2) = \log_5 x^2$

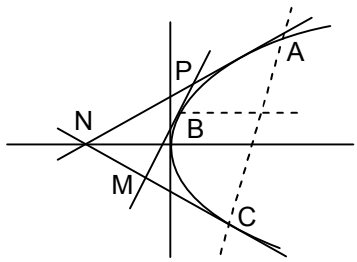
Let $2t = \log_5 x^2 \Rightarrow t > 0$, $x^2 = 5^{2t} \Rightarrow x = 5^t$

$\Rightarrow 5^{3t} - 5^{2t} = 10^{2t} \Rightarrow 5^t - 1 = 4^t \Rightarrow 5^t - 4^t = 1$

$\because$ $5^t - 4^t = 4^t\left(\left(\frac{5}{4}\right)^t - 1\right)$ is an increasing function of t

$\Rightarrow$ This equation has unique solution i.e. t = 1, x = y = z = 5

2. $A(at_1^2, 2at_1)$, $B(at_2^2, 2at_2)$, $C(at_3^2, 2at_3)$



$P(at_1t_2, a(t_1 + t_2))$

$N(at_1t_3, a(t_1 + t_3))$

$M(at_2t_3, a(t_2 + t_3))$

Equation of AC $y(t_1 + t_3) = 2x + 2at_1t_3$

Line through B parallel to x-axis $y = 2at_2$

$L \equiv (a(t_1t_2 + t_2t_3 - t_1t_3), 2at_2)$

Mid-point of L and $N \equiv \left(a\frac{(t_1t_2 + t_2t_3)}{2}, \frac{a(t_1 + 2t_2 + t_3)}{2}\right)$

Mid-point of $PM \equiv \left(a\left(\frac{t_1t_2 + t_2t_3}{2}\right), \frac{a(t_1 + 2t_2 + t_3)}{2}\right)$. Hence, parallelogram

3. For continuity $k = f(0) = \lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{e^{\tan x} - e^x}{\tan x - x} + \lim_{x\to 0} \frac{\ln(\sec x + \tan x) - x}{\left(\frac{\tan x - x}{x^3}\right)\cdot x^3}$

$= \lim_{x\to 0} \frac{e^x\left(e^{\tan x - x} - 1\right)}{\tan x - x} + 3\lim_{x\to 0} \frac{\ln(\sec x + \tan x) - x}{x^3}$

$= 1 + 3\lim_{x\to 0} \frac{\sec x - 1}{3x^2} = 1 + \frac{1}{2} = \frac{3}{2}$

4. $\alpha + \beta + \gamma = 0$, $\alpha\beta\gamma = -\frac{2008}{8}$

Now, let $\alpha + \beta = A$, $\beta + \gamma = B$, $\gamma + \alpha = C$

$\therefore A + B + C = 2(\alpha + \beta + \gamma) = 0$

Hence, $A^3 + B^3 + C^3 = 3ABC$

$\therefore (\alpha + \beta)^2 + (\beta + \gamma)^3 + (\gamma + \alpha)^3 = 3(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha) = -3\alpha\beta\gamma$

5. $N =$ | a | b | c | d |

a can be chosen in 2 ways i.e. 4 or 5

b and c can be chosen in 6 ways i.e. 34, 35, 36, 45, 46, 56

d can be 0 or 5

Hence, total number = $2 \times 6 \times 2 = 24$

6. $^8C_3 \cdot p^3(1-p)^5 = \frac{1}{25}\cdot {}^8C_5 p^5 (1-p)^3$

$\Rightarrow 1 - p = \frac{p}{5} \Rightarrow p = \frac{5}{6}$

7. $\frac{^{10}C_r}{2^r} > \frac{^{10}C_{r-1}}{2^{r-1}}$ and $\frac{^{10}C_r}{2^r} > \frac{^{10}C_{r+1}}{2^{r+1}}$

$\Rightarrow \frac{8}{3} < r < \frac{11}{3}$ $\therefore r = 3$

8. Let G.P. is a, ar, ar^2 .....

Now, $a + ar = 12$ and $ar^2 + ar^3 = 48$

$\Rightarrow r^2 = 4 \Rightarrow r = -2$ and hence $a = -12$

9. $\frac{dx}{dy} = \frac{\frac{dx}{dt}}{\frac{dy}{dt}} = \frac{12t^2}{12t^3} = \frac{1}{t}$

Now, $\frac{d^2x}{dy^2} = \frac{d}{dy}\left(\frac{dx}{dy}\right) = \frac{d}{dt}\left(\frac{1}{t}\right)\frac{dt}{dy} = -\frac{1}{t^2}\frac{1}{12t^3} = -\frac{1}{12t^5}$

$\therefore \frac{\frac{d^2x}{dy^2}}{\left(\frac{dx}{dy}\right)^n} = \frac{-\frac{1}{12t^5}}{\frac{1}{t^n}} = \frac{-t^{n-5}}{12}$ = constant $\Rightarrow n = 5$

10. Write $f(x)=\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^3-3\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)-4\left[\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2-2\right]$

Let $\sqrt{x}+\frac{1}{\sqrt{x}}=t$ $(x>0)$

Let $g(t)=t^3-3t-4t^2+8$

$g'(t)=(t-3)(3t+1)=0\Rightarrow t=3$

$g''(3)>0\Rightarrow g(3)=-10$

11. Tangent at (t, t^3-at) is given by $y=(3t^2-a)x-2t^3$

This line cuts the curve again at $x=-2t$, where slope of tangent is $12t^2-a$

As required $(12t^2-a)(3t^2-a)=-1$

$\Rightarrow 36t^4-15t^2a+a^2+1=0$

For positive root, $a>0$ and discriminant ≥ 0

i.e. $a>0$ and $225a^2-144(a^2+1)=9(3a-4)(3a+4)\geq 0$

$\Rightarrow a\in\left[\frac{4}{3},\infty\right)$

12. We have $\frac{\sin A}{C\sin B}=\frac{a}{bc}$

$\therefore \frac{\sin B}{c}+\frac{\sin C}{b}=\frac{c}{ab}+\frac{b}{ac}\Rightarrow \frac{b\sin B+c\sin C}{bc}=\frac{c^2+b^2}{abc}$

$\Rightarrow a=\frac{b^2+c^2}{b\sin B+c\sin C}=\frac{b(2R\sin B)+c(2R\sin C)}{b\sin B+c\sin C}$

$\Rightarrow a=2R\Rightarrow \angle A=\frac{\pi}{2}$

13. The roots of A are unit orthogonal vectors

$\Rightarrow \vec{R}_1\cdot\vec{R}_2=0\Rightarrow 2b^2=c^2$

$\vec{R}_2\cdot\vec{R}_3=0\Rightarrow a^2-b^2-c^2=0$

$\vec{R}_3\cdot\vec{R}_3=1\Rightarrow a^2+b^2+c^2=1$

$\Rightarrow a^2=\frac{1}{2}$, $b^2=\frac{1}{6}$, $c^2=\frac{1}{3}\Rightarrow |abc|=\frac{1}{6}$

14. $\sum_{i=1}^{n}a_{ii}=a_{11}+a_{22}+.....+a_{nn}$

$$\lim_{n\to\infty}\frac{\sum_{i=1}^{n}a_{ii}}{n^{101}}=\lim_{n\to\infty}\frac{2\left(1^{100}+2^{100}+3^{100}+.....+n^{100}\right)}{n^{101}}=2\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{100}\left(\frac{r}{n}\right)^{100}=2\int_a^1 x^{100}dx=\frac{2}{101}$$

15. We have $a^2+b^2+c^2+ab+bc+ca\leq 0$

$\Rightarrow (a+b)^2+(b+c)^2+(c+a)^2\leq 0$

$\Rightarrow a+b=0, b+c=0, c+a=0$

$\Rightarrow a=b=c$

Now, $\begin{vmatrix}4&0&1\\1&4&0\\0&1&4\end{vmatrix}=65$

16. $|z - (1 + i)|^2 = 2$

$\Rightarrow (x - 1)^2 + (y - 1)^2 = 2$

$\Rightarrow x^2 + y^2 = 2(x + y)$ ..... (1)

Let $w = h + ik = \frac{2}{z} = \frac{2}{x + iy} = \frac{2(x - iy)}{x^2 + y^2}$, so

$h = \frac{2x}{x^2 + y^2}, k = \frac{-2y}{x^2 + y^2}$

$\Rightarrow h - k = \frac{2(x + y)}{x^2 + y^2} = 1$ (from 1)

$\therefore$ Locus of w is $x - y = 1$

17. Point M lies on director circle of hyperbola, hence angle between the tangents is $\frac{\pi}{2}$

as $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{2} = \tan^{-1}1 = \frac{\pi}{4} \Rightarrow 2\left(\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{2}\right) = \frac{\pi}{2}$

18. Let p(a, 4 – a)

The equation of chord of contact AB is

$xa + y(4 - a) = 8$ ..... (1)

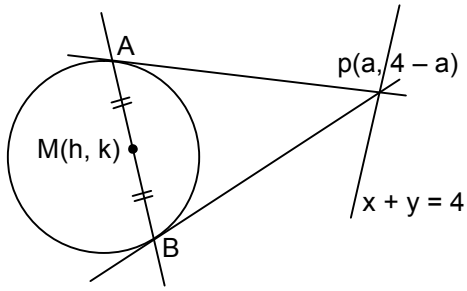
Equation of chord AB with M(h, k) as midpoint

$hx + ky = h^2 + k^2$ ..... (2)

(1) and (2) are same

$\Rightarrow \frac{a}{h} = \frac{4 - a}{k} = \frac{8}{h^2 + k^2}$

$\Rightarrow x^2 + y^2 - 2x - 2y = 0$



19. Circum-centre is (0, 0)

Centroid (G) is $\left(\frac{5 - 13\cos\theta + 13\sin\theta}{3}, \frac{12 + 13\sin\theta - 13\cos\theta}{3}\right)$

Centroid divides line joining orthocentre and circum-centre in ration 2 : 1

$\therefore \frac{5 - 13\cos\theta + 13\sin\theta}{3} = \frac{2 \times 0 + 1 \times h}{3}$ and $\frac{12 + 13\sin\theta - 13\cos\theta}{3} = \frac{2 \times 0 + 1 \times k}{3}$

$\Rightarrow \sin\theta - \cos\theta = \frac{h - 5}{13}$ and $\sin\theta - \cos\theta = \frac{k - 12}{13}$

Hence, locus is $x - y + 7 = 0$

20. Domain of f(x) is x = 1 so, $f(1) = \sqrt{0} + \sin^{-1}\frac{2}{4} = \frac{\pi}{6}$

21. $\lim_{n\to\infty} \frac{n \cdot 3^n}{n(x - 2)^n + n \cdot 3^{n+1} - 3^n} = \frac{1}{3}$, so $\lim_{n\to\infty} \frac{1}{\left(\frac{x - 2}{3}\right)^n + 3 - \frac{1}{n}} = \frac{1}{3}$

$-1 < \frac{x - 2}{3} < 1 \Rightarrow -1 < x < 5$

Hence, possible number of integers in the range = 5

22. $I = \int \left(x^{7m-1} + x^{2m-1} + x^{m-1}\right)\left(2x^{7m} + 7x^{2m} + 14x^m\right)^{1/m} dx$

Put $2x^{7m} + 7x^{2m} + 14x^m = t^m$

$I = \frac{1}{14}\int t^{m-1}\left(t^m\right)^{1/m} dt = \frac{t^{m+1}}{14(m+1)} + c$

23. $F\left(x^2\right) = \int_0^{x^2} t f(t)\,dt = x^4 + x^5$ ..... (1)

On differentiating $2x(x^2)f(x^2) = 4x^3 + 5x^4$

$\Rightarrow f\left(x^2\right) = 2 + \frac{5}{2}x$ ..... (2)

$\therefore \sum_{r=1}^{12} f\left(r^2\right) = \sum_{r=1}^{12}\left(2 + \frac{5}{2}r\right) = 216$

Hence, $\sum_{r=1}^{12} f\left(r^2\right) = 219$

24. On solving $C : y = \frac{1+x}{1+x^2}$ and the line L : y = 1 gives x = 0, 1

Required area = $\int_0^1\left(\frac{x+1}{x^2+1} - 1\right)dx = \frac{1}{2}\ln\left(x^2+1\right) + \tan^{-1}x - x\Big|_0^1 = \frac{1}{2}\ln 2 + \frac{\pi}{4} - 1$

25. $\frac{dx}{dy} = e^y - x \Rightarrow \frac{dx}{dy} + x = e^y$

Hence, integrating factor = e^y

Solution is $xe^y = \frac{1}{2}e^{2y} + c$

$\Rightarrow y = \ln\left(x + \sqrt{x^2+1}\right)$

26. $\left((\vec{a}\times\vec{b})\times\vec{a}\right)\left((\vec{b}\times\vec{a})\times\vec{b}\right) = \left((\vec{a}\cdot\vec{a})\vec{b} - (\vec{b}\cdot\vec{a})\vec{a}\right)\left((\vec{b}\cdot\vec{b})\vec{a} - (\vec{a}\cdot\vec{b})\vec{b}\right)$

$= \left(\vec{a}^2\,\vec{b}^2\right)\vec{a}\cdot\vec{b} - \vec{a}^2\cdot\vec{b}^2(\vec{a}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})\vec{a}^2\cdot\vec{b}^2 + (\vec{a}\cdot\vec{b})^3 = -\left[(\vec{a}\cdot\vec{b})\vec{a}^2\cdot\vec{b}^2 - (\vec{a}\cdot\vec{b})^3\right]$

$= -(\vec{a}\cdot\vec{b})\left[\vec{a}^2\cdot\vec{b}^2 - (\vec{a}\cdot\vec{b})^2\right] = -(\vec{a}\cdot\vec{b})|\vec{a}\times\vec{b}|^2$

27. Line is $\frac{x-2}{1} = \frac{y-1}{1} = \frac{z+\frac{1}{2}}{-\frac{1}{2}} = t$ (Let)

Plane : x + 2y + 6z – 10 = 0

Vector along line : $\vec{V} = \hat{i} + \hat{j} - \frac{1}{2}\hat{k}$

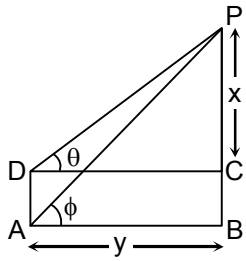
Vector normal to the plane $\vec{n} = \hat{i} + 2\hat{j} + 6\hat{k}$

$\vec{V}\cdot\vec{n} = 0 \Rightarrow$ line is parallel to plane

Distance of $\left(2, 1, -\frac{1}{2}\right)$ from the plane x + 2y + 6z – 10 = 0 is $\frac{9}{\sqrt{41}}$

28. R is reflexive

29. Let AD be the building of height h and BP be the hill.



Then $\tan\phi = \frac{h+x}{y}$ and $\tan\theta = \frac{x}{y}$

$\Rightarrow \tan\phi = \frac{h+x}{x\cot\theta} \Rightarrow x = \frac{h\cot\phi}{\cot\theta - \cot\phi}$

$\Rightarrow h + x = \frac{h\cot\theta}{\cot\theta - \cot\phi}$

30. Mean deviation calculation about mean

x_i	f_i	f_ix_i	$\lvert x_i - 15\rvert$	$f_i\lvert x_i - 15\rvert$
3	8	24	12	96
9	10	90	6	60
17	12	204	2	24
23	9	207	8	72
27	5	135	12	60
$N = \sum f_i = 44$		$\sum f_ix_i = 660$	$\sum f_i\lvert x_i - 15\rvert = 312$	

Mean = $\bar{X} = \frac{1}{N}\left(\sum f_ix_i\right) = \frac{660}{44} = 15$

Mean deviation = M.D. = $\frac{1}{N}\sum f_i\lvert x_i - 15\rvert = \frac{312}{44} = 7.09$